HRISHIKESH BELAGALI

Undergraduate Research Assistant

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SUMMARY

PhD applicant with research interests in quantum algorithms for time evolution, *ab initio* and machine-learning molecular dynamics, fault-tolerant quantum compilation, and quantum error correction.

SKILLS

Languages: Python, Rust, C++, Tcl, LaTeX

Packages: Qiskit, Cirq, NumPy, PyTorch, JAX, SciPy

Software: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S.) — **Michigan State University Honors College**
2024–2027 (expected)
Current GPA: 4.0

Relevant coursework: Quantum Algorithms†, Tensor Networks†, Quantum Physics, Computational Foundations of Artificial Intelligence†, Linear Algebra
(† Graduate course)

EXPERIENCE

2026–Present **Undergraduate Research Assistant** — **MSU-Q** (Advisor: Prof. Ryan LaRose)
- Developed fermionic backpropagation, an efficient classical method for simulating large-scale unitary cluster Jastrow circuits, combining new analytical reductions with a GPU-accelerated implementation to achieve a 70-fold speedup for circuits with up to 77 qubits and 3,500 two-qubit gates.
- Built *propaq*, a Heisenberg propagation package achieving runtimes competitive with state-of-the-art simulators while offering greater customizability, enabling circuit simulations in arbitrary bases with support for custom noise and truncation policies.
- Investigated unitary basis transformations for improving error correction code performance in the presence of biased noise.
- Analyzed the use of the canonical polyadic decomposition for pre-processing chemistry Hamiltonians to reduce time evolution costs.

2025–2026 **Honors Research Scholar** — **MSU Honors College/MSU-Q** (Advisor: Prof. Ryan LaRose)
- Developed a gateset-rotation algorithm for identifying unitary basis transformations to reduce non-Clifford synthesis cost for fault-tolerant quantum algorithms.
- Investigated the utility of unitary basis transformations for Hamiltonian pre-processing in time evolution algorithms.

2024–Present **Undergraduate Research Assistant** — **Vermaas Lab, Plant Research Laboratory** (Advisor: Prof. Josh Vermaas)
- Simulated the effects of butanol and isobutanol on *Clostridium acetobutylicum* and *Zymomonas mobilis* membranes to characterize their impact on membrane dynamics.
- Ran large-scale molecular dynamics simulations on high-performance computing clusters (40,000+ GPU hours and 30TB+ of data) using NAMD.
- Identified key membrane components contributing to alcohol tolerance through data analysis in Python, Tcl, and VMD.
- Investigated the effect of the PDLP5 protein cap on water transport in PIP1;4 aquaporins via grid-steered molecular dynamics.

PUBLICATIONS

[1] **Hrishikesh Belagali**, Ryan LaRose, "Propaq: A Python package for Heisenberg Propagation," *arXiv:2609.07730 (2026)*. Submitted to *ACM Transactions on Software Engineering and Methodology*.

[2] **Hrishikesh Belagali**, Thomas Van Camp, R. Pradeep, Sourin Das, Namit Anand, Ryan LaRose, "Efficient classical simulation of large-scale unitary cluster Jastrow circuits," *arXiv:2607.21337 (2026)*.
Submitted to *Physical Review Letters* (revised manuscript under review).

[3] **Hrishikesh Belagali**, Nitin Kumar Singh, Josh V. Vermaas, "The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*," ChemRxiv:15004463 (2026).
Submitted to *The Journal of Physical Chemistry B* (under revision).

CONFERENCES

2026
ACS, IL

Molecular mechanisms of alcohol tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*: The role of cyclopropane fatty acids and hopanoids — **acs.digitellinc.com**
Poster presented at ACS Fall 2026 Conference.

2026 *MQC, IN*

Efficient classical simulation of large-scale unitary cluster Jastrow circuits
Poster presented at MQC Entanglement 2026.

2026 *GLBRC, WI*

The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*
Poster presented at Great Lakes Bioenergy Research Center Annual Science Meeting.

2026 *MSU UURAF*

Reducing T Gate Counts via Gateset Rotation
Poster presented at MSU Undergraduate Research and Arts Forum. **Awarded first place** in the Engineering, Computer Science and Mathematics division.

2025 *ACS, D.C.*

Aquaporin Water Transit Modulation via Protein Capping **acs.digitellinc.com**
Poster presented at ACS Fall 2025 Conference.

2025 *MI*

Evaluating the Interaction of Chloroplast-targeting Protein (At4g16060) with Thylakoid Membrane
Poster presented at First Great Lakes Plant Science Conference.

RESEARCH SOFTWARE & OPEN SOURCE

propaq
Python, Rust

propaq **github.com/pypi.org**
Creator and primary developer of *propaq*, a Heisenberg propagation library for quantum circuit simulation.
- Implemented thread-safe kernels and compact representation of Pauli and Majorana strings in Rust for efficient multi-threaded simulation.
- Developed PyO3-based Python bindings integrating *propaq* with Qiskit and Cirq for seamless workflows.
- Built a low-overhead plugin system for custom noise/truncation policies in Python, Rust, C, and Julia.

ffsim
Python, Rust

ffsim **PR #684**
Contributed to *ffsim*, a Python package for simulating fermionic quantum systems.
- Implemented fermionic backpropagation (arXiv:2607.21337) in *ffsim* for fast UCJ ansatz simulation.
- Created a JAX-based implementation for GPU-accelerated variational optimization.

QuTiP
Python

Quantum Toolbox in Python (QuTiP) **PR #2731**
Contributed to QuTiP: added support for returning eigenkets as a Qobj operator.

aqua-blue
Python

aqua-blue **github.com/pypi.org**
Secondary maintainer of aqua-blue, a Python package for building echo state networks.
- Implemented reservoir state time-lag, sparsity, and spectral radius to improve prediction accuracy.
- Introduced timezone-aware datetime support for seamless interfacing with data pipelines.
- Demonstrated forecasting capability on the quantum time evolution of Gaussian wave packets.

tce-lib
Python

tce-lib **PR #77**
Contributed to tce-lib, a Python package for tensorial cluster expansion models of concentrated alloys.
- Implemented lazy evaluation of Monte Carlo trajectories for efficient simulation on HPC clusters.

HONORS AND AWARDS

- First place at MSU UURAF in the Engineering, Computer Science and Mathematics division (2026)
- Wielenga Research Scholarship, Honors College (2025–2026) – Awarded to 15–20 MSU students per year
- Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024) – $2,000
- MSU President's Scholarship (2024) – $30,000

RESEARCH COMMUNICATION

2026 *MSU CoE, MI*

"Student View: Challenging the Quantum Advantage Boundary" **engineering.msu.edu**
Article on my quantum computing research journey, published by the MSU College of Engineering.

SELECTED PROJECTS

Python
Qiskit

Adiabatic Quantum Computation **github.com**
Trotterized adiabatic algorithm for QUBO-type problems via Ising formulations; extended in my Graduate Quantum Algorithms course to study hardware-noise effects on convergence.

Python
OVITO, PyMOL

Genetic Annealing to Determine Protein Structures **github.com**
Monte Carlo Genetic Algorithm to determine protein structures through the optimization of Irbäck's off-lattice model energy equation (RMSD < 3.0 Å).